

Yanzhen Chen

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EDUCATION

University of Toronto

September 2023 – Expected April 2028

Honours Bachelor of Science in Computer Science (Co-op) | CS GPA: 3.82

Toronto, ON

Relevant Coursework: Data Structures and Algorithms, Operating Systems, Databases, Deep Learning

Teaching Assistant: CSC165: Mathematical Expression and Reasoning for Computer Science (Winter 2026)

EXPERIENCE

Bree (YC S21)

Toronto, ON

Software Engineer Intern

May 2026 – Present

- Leading the migration of Bree's production **React Native** application from React Navigation to **Expo Router** across **78 screens and 230+ call sites**; scoped the initiative into a **4-milestone, 25-ticket roadmap** and coordinated incremental production releases.
- Designed a backward-compatible routing architecture supporting simultaneous legacy and new navigation, with **typed routes, deep linking, analytics instrumentation, and automated route validation** while preserving compatibility across the application's **745-test suite**.
- Shipped full-stack features for a fintech platform serving **800,000+ users** using **React, TypeScript, AWS Lambda, and PostgreSQL**, including secure data exports, workflow automation, and internal tooling for customer operations.

University of Toronto, MarkUs Project

Toronto, ON

Open Source Contributor

May 2026 – Present

- Landed **20 merged pull requests** to MarkUs, an open-source **Ruby on Rails** assignment-management platform used by **15,000+ students** across the University of Toronto, University of Waterloo, and École Centrale de Nantes, improving authentication, instructor workflows, course administration, and testing infrastructure.

IQBank

Toronto, ON

Technical Lead & Software Engineer

January 2026 – Present

- Leading a team of **6 developers**, driving architecture decisions, reviewing pull requests, mentoring contributors, and coordinating frontend and backend development.
- Built full-stack features with **React, TypeScript, Node.js, and PostgreSQL**, including role-based access control, persistent drafts, user profiles, and bulk account creation for an exam-preparation platform serving **300+ students per semester**.

PROJECTS

NovaOS | TypeScript, React, Vite, Vitest, Turborepo | Demo

June 2026 – Present

- Built a browser-based operating systems laboratory as a **16-package TypeScript monorepo**, including a C-like compiler performing parsing, semantic analysis, optimization, and source-mapped code generation.
- Designed a custom **32-bit virtual machine and microkernel** with **30 instructions**, process scheduling, system calls, memory management, and inter-process communication.
- Built a **time-travel debugger** supporting deterministic replay, source-level stepping, **5 breakpoint types**, call-stack reconstruction, and live register and memory inspection.

TorontoGuessr | TypeScript, Next.js, Tailwind, Supabase, Google Maps API | Demo

April 2025 – Present

- Built and deployed a full-stack geolocation game with **Next.js, PostgreSQL**, and **Google Street View**, serving **300+ monthly players** with more than **4,000 games played**.
- Designed APIs for randomized Street View generation, persistent scoring, and leaderboards backed by more than **2,000 curated Toronto locations**.

LEADERSHIP EXPERIENCE

President

May 2026 – Present

Computer Science Student Union

Toronto, ON

- Leading a **10-person executive team** serving **2,300+ computer science students**; previously served two years as VP External Relations, securing **\$14,300 in sponsorships** and industry partnerships.

TECHNICAL SKILLS

Languages: Python, TypeScript, JavaScript, Java, C++, C, Go, Ruby, SQL

Frameworks: React, React Native, Next.js, Node.js, Express, Expo, Ruby on Rails

Infrastructure & Tools: AWS, Docker, PostgreSQL, Supabase, Prisma, Linux/Unix, Git, GitHub Actions, Claude Code